

Value Degradation Identification Module (VDGIM)

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Canonical Definition

Value Degradation Identification Module (VDGIM) is the first Core Module of the Value Degradation Standard (VDGS) within VFM® — Value Flow Mechanisms Architecture.

VDGIM governs the identification of whether existing positive value has entered a materially negative developmental condition characterized by weakening, decline, deterioration, erosion, impairment, loss of capability, loss of utility, or another context-relevant reduction while positive value continues to exist.

The Module establishes the first governed determination that a potential degradation signal represents a legitimate Value Degradation Candidate requiring further degradation governance.

VDGIM separates actual degradation from ordinary variation, temporary fluctuation, neutral Value Change, measurement noise, contextual reclassification, and Value Nullification.

Its foundational task is:

to determine whether degradation is actually occurring before its condition, magnitude, progression, or downstream consequences are governed.

Core Question

Has existing positive value entered a materially degrading condition, and what exactly is degrading?

Architectural Position

VDGIM is the first module within VDGS.

It receives governed value information from the preceding VFM® lifecycle, particularly:

VSS — Value State Standard

VCS — Value Change Standard

and, where relevant:

VDS — Value Development Standard.

VDGIM establishes whether an observed negative development should enter Value Degradation governance.

It precedes:

VDGCM — Value Degradation Condition Module.

The internal sequence begins:

Existing Value → Degradation Identification → Degradation Condition.

VFM® Governance Flow Position

Value Context → Value Origin → Value Emergence → Value Recognition → Value State → Value Change → Value Development → Value Degradation → subsequent VFM® lifecycle spaces

VDGIM operates at the entry point into the eighth governed space:

Value Degradation.

Its function is to determine whether the value has legitimately crossed from:

existing value experiencing ordinary change

into:

existing value experiencing degradation.

Internal Module Flow

Existing Positive Value → Potential Degradation Signal → Reference Value Condition Identification → Affected Value Dimension Identification → Negative Change Confirmation → Materiality Assessment → Persistence / Recurrence Assessment → Alternative Explanation Review

→ Remaining Positive Value Confirmation

→ Value Degradation Identification → VDGCM — Value Degradation Condition Module

The logic is:

Is there a negative signal? → What value is affected? → Compared with what baseline? → Is the change material? →

Is it actually degradation? → Does positive value remain? → What degradation condition should now be examined?

Operational Space

VDGIM governs the point at which an observed deterioration signal becomes an identified Value Degradation condition requiring further governance.

Its operational space includes:

degradation candidate identification; governed value object identification; reference value condition identification; affected value dimension identification; negative change detection; deterioration signal identification; materiality screening; persistence screening; recurrence screening; baseline comparison; degradation boundary identification; alternative explanation review; temporary variation exclusion; measurement anomaly exclusion; neutral change exclusion; Value Development distinction; Value Nullification distinction; remaining positive value confirmation; degradation identification state; degradation identification record.

Value Degradation Candidate

A Value Degradation Candidate is:

a governed value condition, observation, signal, measurement, event, pattern, or change indicating that existing positive value may be materially weakening, declining, deteriorating, eroding, or becoming impaired.

A candidate does not yet establish confirmed degradation.

It creates the governed question:

Has degradation actually occurred?

Potential degradation signals may include:

reduced productive capability; declining performance; increasing failure frequency; reduced reliability; declining utility; physical deterioration; technological obsolescence;

knowledge decay; declining relevance; reduced interoperability; increased operating burden; loss of efficiency; reduced demand; capability erosion; network weakening; declining strategic utility; repeated negative Value Changes; another materially relevant deterioration signal.

Existing Value Principle

VDGIM applies only where value already exists.

The Module therefore requires an identifiable existing positive value basis.

Conceptually:

Value Exists

before:

Value Degrades.

This preserves the VFM® lifecycle distinction:

Value Emergence ≠ Value Degradation.

Potential value that never emerged cannot be described as degraded existing value merely because its expected potential was not realized.

Degradation Object Identification

VDGIM must identify what value is allegedly degrading.

The degradation object may be:

an asset; capability; system; infrastructure; technology; knowledge object; intellectual asset; relationship; network; dataset; product; service; organization; process; productive capacity; strategic position; economic position; another governed value-bearing object.

The object must be sufficiently bounded to prevent degradation observed in one component from being automatically generalized to an entire system.

Reference Value Condition

Degradation cannot be meaningfully identified without a reference condition.

VDGIM therefore establishes the relevant Reference Value Condition.

This may be:

Prior Governed Value State

Recognized Value Condition

Post-Development Value Condition

Expected Maintained Condition

Established Performance Baseline

Previous Capability Condition

Previous Utility Condition

or another legitimate governed reference.

The comparison becomes:

Reference Value Condition → Current Value Condition.

Without an appropriate reference, a claim of degradation may remain methodologically ambiguous.

Negative Change Confirmation

VDGIM determines whether the relevant value dimension has actually moved negatively relative to the governed reference.

The Module may identify:

Value at T_1 < Value at T_0

where quantitative representation is legitimate.

But numerical representation is not mandatory.

Negative change may also be established through:

capability reduction; condition deterioration; utility loss; performance decline; relevance loss; reliability reduction; structural impairment; qualitative deterioration; another context-appropriate basis.

Negative Change ≠ Value Degradation

A negative Value Change is an important input.

It is not sufficient by itself to establish degradation.

For example:

A temporary reduction in market price may not represent degradation of the underlying productive capability.

A temporary production interruption may not represent deterioration of the production asset.

A temporary decline in usage may not establish deterioration of the underlying service capability.

VDGIM therefore requires:

Negative Change + Material Degradation Basis → Value Degradation Identification

not:

Any Negative Change → Value Degradation.

Materiality Principle

VDGIM requires the observed deterioration to be materially relevant within the defined Value Context.

Materiality may depend upon:

magnitude; duration; recurrence; persistence; affected value dimension; operational significance; economic significance; strategic significance; capability significance; relationship to the governed purpose; another context-specific factor.

VFM® does not impose one universal degradation threshold.

The threshold must be appropriate to the governed value.

Temporary Variation Boundary

Value may temporarily move downward without entering a degradation pathway.

VDGIM therefore distinguishes:

Temporary Negative Variation

from:

Value Degradation.

Examples may include:

seasonal demand reduction; temporary market volatility; short maintenance shutdown; temporary resource constraint; transient system performance reduction; temporary network interruption; reversible short-term operational disruption.

The existence of recovery after a short variation may provide evidence that the condition was not persistent degradation.

However, reversibility alone does not exclude degradation.

A genuinely degraded value may also later be restored.

Persistence and Recurrence

Persistence and recurrence can strengthen degradation identification.

A negative condition that:

persists

repeats

accumulates

or

intensifies

may indicate that the value has entered a degradation pathway.

VDGIM therefore asks:

Is this a momentary condition, or is a deteriorating pattern emerging?

The answer may affect whether the candidate is:

Unconfirmed

Potential

or

Identified Degradation.

Measurement Noise Boundary

Not every observed decline represents a real decline in value.

Differences may arise from:

measurement error; inconsistent methodology; changed measurement basis; data quality issues; sampling variation; reporting differences; incomplete observation; model changes; classification changes.

VDGIM therefore requires material measurement uncertainty to remain visible.

Where the apparent degradation cannot be distinguished from measurement uncertainty, the identification should remain unresolved rather than being presented as confirmed degradation.

Context Change Boundary

The Value Context itself may change.

A value object may appear less valuable under one context without the underlying value-bearing capability having deteriorated.

For example:

A machine may retain identical productive capability while demand for its output declines.

A dataset may remain technically intact while the organizational purpose for which it was collected changes.

A technology may remain fully functional while a new technology changes its competitive position.

Depending on the governed value dimension, these conditions may represent genuine degradation, contextual revaluation, or both.

VDGIM therefore requires the affected value dimension to be explicit.

Value Dimension Identification

Value Degradation must be attached to the dimension in which deterioration occurs.

Relevant dimensions may include:

Economic Value

Operational Value

Functional Value

Technical Value

Strategic Value

Knowledge Value

Relational Value

Network Value

Productive Value

Reliability Value

Utility Value

Physical Value

or another governed dimension.

The Module must avoid the uncontrolled inference:

One Dimension Degraded → All Value Degraded.

Multidimensional Identification

A value-bearing object may simultaneously exhibit:

Operational Value ↓

Economic Value ↓

Knowledge Value →

Strategic Value ↑

VDGIM may therefore identify multiple degradation candidates independently.

This allows VFM® to preserve the real structure of value rather than forcing complex value into a single aggregate state.

Degradation Identification State

VDGIM may establish the following minimum identification states:

1. No Degradation Identified

Available information does not establish a material degradation condition.

2. Potential Degradation

A deterioration signal exists but the basis is insufficient for confirmed identification.

3. Degradation Identified

Existing positive value has been determined to be materially degrading within the defined value dimension and context.

4. Degradation Identification Suspended

A prior identification requires reassessment because the baseline, measurement basis, context, evidence, or value condition has materially changed.

5. Degradation Identification Withdrawn

A previous identification is no longer supported by the governed basis.

These states prevent uncertainty from being hidden behind a binary yes/no classification.

Remaining Positive Value Confirmation

VDGIM must preserve the defining VDGS boundary:

positive value still exists.

If the relevant governed value has reached zero, the condition is no longer merely Value Degradation within that value dimension.

The lifecycle has reached the boundary toward Value Nullification.

Therefore:

Positive Value → Lower Positive Value = Degradation Candidate

while:

Positive Value → Zero Value = Nullification Boundary

where applicable.

Severe Degradation ≠ Zero Value

The magnitude of deterioration does not itself determine nullification.

An asset may lose:

20%

50%

90%

or even more of a particular value dimension while still retaining positive value.

VDGIM therefore requires remaining value to be considered explicitly.

This prevents severely degraded assets from being prematurely classified as ZVA™ — Zero Value Asset™.

Value Development Boundary

Value Development and Value Degradation may occur across different dimensions at the same time.

For example:

A legacy technology may experience:

Commercial Value ↓

while:

Historical Knowledge Value ↑

VDGIM must therefore identify degradation without erasing simultaneous development elsewhere.

The correct representation may be:

Development in Dimension A

External Cause Boundary

VDGIM may identify an external event associated with the degradation signal.

Such events may include:

market change; technological development; physical deterioration; regulation; payment event; environmental change; organizational change; operational event; competitive event; human action; another external mechanism.

VDGIM does not need to determine the complete causal explanation.

Its primary question remains:

Has the governed value materially degraded?

Detailed governance of the degradation condition follows under VDGCM.

Evidence Basis

VDGIM may use:

measurements; performance records; condition observations; operational data; financial data; market data; maintenance records; reliability records; technical assessments; user behavior; utilization records; historical comparison; expert assessment; validated evidence; another legitimate basis.

The evidence basis must be proportionate to the significance of the degradation determination.

Temporal Identification

Degradation identification must remain temporally anchored.

Where relevant, VDGIM should establish:

T_0 — Reference Value Condition

→ T_1 — Initial Negative Signal

→ T_2 — Materiality / Persistence Observation

→ T_3 — Degradation Identification

This allows the organization to distinguish when deterioration began from when it was formally recognized.

Degradation Start $\neq$ Degradation Identification Date.

Retrospective Identification

VDGIM permits degradation to be identified retrospectively.

An organization may discover later that deterioration began earlier than originally recognized.

For example:

A machine may show increasing defect rates for six months before the pattern is formally identified as degradation.

The governed record may therefore distinguish:

Estimated Degradation Onset

from:

Identification Confidence

Where appropriate, VDGIM may record confidence associated with the degradation identification.

Confidence may be:

high; moderate;

limited; provisional; another governed state.

Confidence must not replace the identification basis.

Its role is to communicate uncertainty concerning the determination.

VDGIM Boundary with VCS

VCS governs:

what materially changed in value.

VDGIM governs:

whether a negative change constitutes identifiable Value Degradation.

Therefore:

Value Change → possible degradation signal

but:

Value Change ≠ Value Degradation Identification.

VDGIM Boundary with VSS

VSS governs the current governed Value State.

VDGIM identifies whether that value has entered a degradation pathway.

Therefore an object may simultaneously be:

Value State: Positive Value Asset

and

Degradation Identification: Degradation Identified.

VDGIM Boundary with VDS

VDS governs positive development of existing value.

VDGIM governs identification of negative development.

Neither should erase the other where different value dimensions move in different directions.

VDGIM Boundary with VDGCM

VDGIM answers:

Is Value Degradation occurring?

VDGCM answers:

Under what degradation condition is that value deteriorating?

Therefore:

Identification → Condition.

VDGIM must not absorb the detailed degradation-condition governance owned by VDGCM.

VDGIM Boundary with Value Nullification

VDGIM operates while positive value remains.

Where the governed value reaches zero, the appropriate downstream VFM® mechanism must govern the zero-value transition and resulting state.

Therefore:

Degradation Identification ≠ Nullification Determination.

VDGIM Boundary with External Architectures

VDGIM does not replace:

VALIDOS® for validation governance.

INTEGROS® for integrity governance.

ADIT® for auditability governance.

ORA™ for Operational Reality governance.

AGA™ for accountability governance.

These architectures may provide governed inputs to VDGIM.

VDGIM retains the narrower responsibility of identifying Value Degradation.

Minimum Implementation Framework

A minimum VDGIM implementation should establish:

1. Governed Value Identification

Identify the existing value-bearing object and relevant value dimension.

2. Reference Condition

Establish the legitimate baseline or previous governed value condition.

3. Degradation Signal

Identify the observation, event, pattern, measurement, or change indicating potential deterioration.

4. Negative Change Confirmation

Determine whether value has moved negatively relative to the reference condition.

5. Materiality Assessment

Determine whether the negative movement is materially relevant.

6. Persistence / Recurrence Assessment

Determine whether the condition is temporary, persistent, recurring, cumulative, or otherwise degradation-relevant.

7. Alternative Explanation Review

Consider measurement noise, context change, reclassification, temporary variation, or another explanation.

8. Remaining Value Confirmation

Confirm whether positive value continues to exist.

9. Identification State

Assign the governed Value Degradation Identification State.

10. Identification Record

Document the value object, dimension, reference condition, signal, basis, timing, uncertainty, and determination.

Governance Outputs

VDGIM may produce:

Value Degradation Candidate Record Value Degradation Object Record Reference Value Condition Record Value Degradation Signal Record Affected Value Dimension Record Negative Value Change Confirmation Degradation Materiality Assessment Persistence / Recurrence Record Alternative Explanation Record Remaining Positive Value Confirmation Value Degradation Identification State Value Degradation Identification Record

These outputs provide the governed input basis for:

VDGCM — Value Degradation Condition Module.

Use Case 1 — Industrial Asset Degradation

A production machine remains operational and continues generating economically useful output.

Over time, however:

cycle time increases

→ energy consumption rises

→ maintenance interventions become more frequent

→ precision deteriorates

→ unplanned downtime increases.

No single observation necessarily proves degradation.

VDGIM establishes the machine's prior governed performance condition, identifies the affected value dimensions, compares current performance with the baseline, tests whether the changes are material and persistent, and separates temporary operational disturbances from a genuine deterioration pattern.

The result may be:

Operational Value: Degradation Identified

Reliability Value: Degradation Identified

while:

Remaining Productive Value: Positive.

The machine therefore remains a value-bearing asset while entering governed Value Degradation.

The next question passes to VDGCIM:

Under what conditions is this degradation occurring?

Use Case 2 — Knowledge and Technology

Degradation

An organization owns a technical knowledge base and software capability that remain operationally usable.

Over several years:

technical dependencies become obsolete

→ integration compatibility declines

→ specialist knowledge leaves the organization

→ documentation becomes outdated

→ maintenance becomes slower

→ new systems increasingly bypass the capability.

The underlying software and knowledge have not disappeared.

Nor has their value reached zero.

VDGIM identifies separate degradation dimensions:

Interoperability Value ↓

Operational Utility ↓

Knowledge Accessibility ↓

Maintenance Capability ↓

while potentially preserving:

Historical Knowledge Value →

or even:

Strategic Legacy Knowledge Value ↑.

The Module prevents the organization from treating the entire asset as either simply valuable or worthless.

Instead, it identifies precisely which value is degrading while positive value continues to exist.

Enterprise Application

VDGIM provides organizations with an early governance control for detecting Value Degradation before deterioration becomes severe or reaches zero.

It can support:

asset management; infrastructure governance; technology lifecycle management; knowledge governance; portfolio governance; capability management; software lifecycle management; network governance; strategic asset governance; operational performance governance; organizational capability assessment.

Its practical purpose is to transform:

“Something appears to be getting worse.”

into:

“A defined value dimension of a defined value-bearing object has entered an identified degradation condition relative to a governed reference basis.”

That distinction makes degradation traceable, comparable, governable, and ready for subsequent condition, magnitude, progression, and record governance.

Foundational Principles

Value must exist before it can degrade.

A degradation signal is not yet a degradation determination.

Negative Value Change is not automatically Value Degradation.

Temporary variation is not automatically degradation.

Measurement difference is not automatically value deterioration.

Degradation must be tied to a defined value object and value dimension.

A legitimate reference condition is required to identify deterioration.

One degraded value dimension does not automatically mean that all value has degraded.

Value Development and Value Degradation may coexist across different dimensions.

Severe degradation does not automatically establish zero value.

Value Degradation remains within VDGS while positive value continues to exist.

Canonical Closing Statement

Value Degradation Identification Module establishes the entry control point of the Value Degradation Standard by determining whether existing positive value has legitimately entered a materially negative developmental condition. Through governance of degradation candidates, value objects, reference conditions, affected dimensions, negative change, materiality, persistence, recurrence, alternative explanations, temporal identification, uncertainty, and remaining positive value, VDGIM prevents ordinary fluctuation, measurement noise, contextual change, neutral Value Change, Value Development, and Value Nullification from being incorrectly classified as Value Degradation. By establishing precisely what value is degrading before its condition, magnitude, progression, or downstream lifecycle consequences are governed, VDGIM creates the methodological foundation for the complete VDGS degradation sequence: degradation must first be identified before it can legitimately be characterized, measured, tracked, or acted upon.

Modul2

Governed Links

→ [Value Degradation Standard \(VDGS\)](#)

→ [VFM® Architecture Map](#)

→ [Next: Value Degradation Condition Module \(VDGCM\)](#)

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